

Solving Equations Practice

Mastering One-Step, Two-Step & Multi-Step Equations

This practice workbook provides extensive drill and practice on solving equations. Work through the examples, then complete the practice problems to build speed and confidence.

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Empowering Students Through Mathematics
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Section 1: One-Step Equations

A **one-step equation** can be solved in a single step using the inverse operation. The goal is to get the variable alone on one side of the equals sign.

Types of One-Step Equations:

- Addition equations ($x + a = b$) → subtract a from both sides
- Subtraction equations ($x - a = b$) → add a to both sides
- Multiplication equations ($ax = b$) → divide both sides by a
- Division equations ($x/a = b$) → multiply both sides by a

Worked Examples

Example 1: Solve: $x + 9 = 14$

$$x + 9 - 9 = 14 - 9 \quad x = 5 \quad \text{Check: } 5 + 9 = 14 \quad \checkmark$$

Example 2: Solve: $y - 7 = 3$

$$y - 7 + 7 = 3 + 7 \quad y = 10 \quad \text{Check: } 10 - 7 = 3 \quad \checkmark$$

Example 3: Solve: $6n = 42$

$$6n \div 6 = 42 \div 6 \quad n = 7 \quad \text{Check: } 6(7) = 42 \quad \checkmark$$

Example 4: Solve: $x/8 = 5$

$$(x/8) \times 8 = 5 \times 8 \quad x = 40 \quad \text{Check: } 40/8 = 5 \quad \checkmark$$

Example 5: Solve: $-4a = 24$

$$-4a \div (-4) = 24 \div (-4) \quad a = -6 \quad \text{Check: } -4(-6) = 24 \quad \checkmark$$

Example 6: Solve: $m + 2.5 = 7$

$$m = 7 - 2.5 \quad m = 4.5 \quad \text{Check: } 4.5 + 2.5 = 7 \quad \checkmark$$

Example 7: Solve: $x - (-3) = 10$

$$x + 3 = 10 \quad x = 7 \quad \text{Check: } 7 - (-3) = 7 + 3 = 10 \quad \checkmark$$

Example 8: Solve: $(\frac{2}{3})x = 8$

$$x = 8 \times (\frac{3}{2}) \quad x = 12 \quad \text{Check: } (\frac{2}{3})(12) = 8 \quad \checkmark$$

Section 2: Two-Step Equations

A **two-step equation** requires two operations to isolate the variable. The general form is: $ax + b = c$ or $ax - b = c$.

Strategy (Reverse PEMDAS):

Step 1: Undo addition or subtraction FIRST (move the constant)

Step 2: Undo multiplication or division SECOND (isolate the variable)

Step 3: Check by substituting your answer back in

Worked Examples

Example 1: Solve: $2x + 5 = 13$

$$2x + 5 - 5 = 13 - 5 \quad 2x = 8 \quad x = 4 \quad \text{Check: } 2(4) + 5 = 13 \quad \checkmark$$

Example 2: Solve: $3y - 8 = 7$

$$3y - 8 + 8 = 7 + 8 \quad 3y = 15 \quad y = 5 \quad \text{Check: } 3(5) - 8 = 7 \quad \checkmark$$

Example 3: Solve: $x/4 + 6 = 10$

$$x/4 = 4 \quad x = 16 \quad \text{Check: } 16/4 + 6 = 10 \quad \checkmark$$

Example 4: Solve: $-5n + 3 = -12$

$$-5n = -15 \quad n = 3 \quad \text{Check: } -5(3) + 3 = -12 \quad \checkmark$$

Example 5: Solve: $7 - 2x = 1$

$$-2x = -6 \quad x = 3 \quad \text{Check: } 7 - 2(3) = 1 \quad \checkmark$$

Example 6: Solve: $(x+3)/2 = 5$

$$x + 3 = 10 \quad x = 7 \quad \text{Check: } (7+3)/2 = 5 \quad \checkmark$$

Example 7: Solve: $4a + 1.5 = 9.5$

$$4a = 8 \quad a = 2 \quad \text{Check: } 4(2)+1.5 = 9.5 \quad \checkmark$$

Example 8: Solve: $-3x - 4 = 8$

$$-3x = 12 \quad x = -4 \quad \text{Check: } -3(-4)-4 = 8 \quad \checkmark$$

Section 3: Multi-Step Equations

Multi-step equations involve more than two steps. You may need to distribute, combine like terms, and/or move variables to one side.

Complete Strategy:

1. Simplify each side (distribute, combine like terms)
2. Get all variable terms on one side (add/subtract)
3. Get all constants on the other side
4. Divide to isolate the variable
5. Always check your solution!

Worked Examples

Example 1: Solve: $3(x + 4) = 21$

$$3x + 12 = 21 \quad 3x = 9 \quad x = 3 \quad \text{Check: } 3(3+4) = 3(7) = 21 \quad \checkmark$$

Example 2: Solve: $2(3x - 1) + 4 = 16$

$$6x - 2 + 4 = 16 \quad 6x + 2 = 16 \quad 6x = 14 \quad x = 7/3 \quad \text{Check: } 2(3 \cdot 7/3 - 1) + 4 = 2(6) + 4 = 16 \quad \checkmark$$

Example 3: Solve: $5x + 3 = 3x + 11$

$$5x - 3x = 11 - 3 \quad 2x = 8 \quad x = 4 \quad \text{Check: } 5(4) + 3 = 23, \quad 3(4) + 11 = 23 \quad \checkmark$$

Example 4: Solve: $4(x - 2) = 2(x + 3)$

$$4x - 8 = 2x + 6 \quad 2x = 14 \quad x = 7 \quad \text{Check: } 4(5) = 20, \quad 2(10) = 20 \quad \checkmark$$

Example 5: Solve: $3x + 2(x - 4) = x + 8$

$$3x + 2x - 8 = x + 8 \quad 5x - 8 = x + 8 \quad 4x = 16 \quad x = 4 \quad \text{Check: } 3(4) + 2(0) = 12, \quad 4 + 8 = 12 \quad \checkmark$$

Example 6: Solve: $7 - 3(2x - 1) = -8$

$$7 - 6x + 3 = -8 \quad 10 - 6x = -8 \quad -6x = -18 \quad x = 3 \quad \text{Check: } 7 - 3(5) = 7 - 15 = -8 \quad \checkmark$$

Section 4: Word Problems with Equations

Word problems require translating English into mathematical equations. Follow these steps:

1. Read the problem carefully — identify what you're solving for
2. Define your variable (Let $x = \dots$)
3. Write an equation based on the problem
4. Solve the equation
5. Check: does your answer make sense in context?

Worked Examples

Example 1:

A number tripled and then decreased by 5 equals 16. Find the number.

$$\begin{aligned} \text{Let } x = \text{the number} \quad & 3x - 5 = 16 \quad 3x = 21 \quad x = 7 \quad \text{Check: } 3(7) \\ & - 5 = 21 - 5 = 16 \quad \checkmark \end{aligned}$$

Example 2:

The perimeter of a rectangle is 42 cm. The length is 3 cm more than the width. Find the dimensions.

$$\begin{aligned} \text{Let } w = \text{width, then length} = w + 3 \quad & 2(w) + 2(w+3) = 42 \quad 2w \\ & + 2w + 6 = 42 \quad 4w = 36 \quad w = 9 \quad \text{Width} = 9 \text{ cm, Length} = 12 \text{ cm} \end{aligned}$$

Example 3:

Movie tickets cost \$9 for adults and \$6 for children. A family of 5 spent \$36. How many adults went?

Let a = adults, then children = $5 - a$
 $9a + 6(5-a) = 36$
 $9a + 30 - 6a = 36$
 $3a = 6$
 $a = 2$
2 adults, 3 children

Example 4:

A phone plan costs \$25/month plus \$0.10 per text. Last month's bill was \$37. How many texts were sent?

Let t = number of texts
 $25 + 0.10t = 37$
 $0.10t = 12$
 $t = 120$ texts

Example 5:

Two consecutive odd integers have a sum of 76. Find them.

Let n = first odd integer, $n+2$ = next
 $n + (n + 2) = 76$
 $2n + 2 = 76$
 $2n = 74$
 $n = 37$
The integers are 37 and 39.

Section 5: Practice Drill — 40 Equations

Directions: Solve each equation. Show all work.

One-Step Equations (1-12)

1. $x + 8 = 15$

2. $y - 6 = 9$

3. $5n = 35$

4. $a/4 = 6$

5. $x + 12 = 7$

6. $m - 9 = -3$

7. $-3x = 21$

8. $y/7 = 8$

9. $x + 3.5 = 10$

10. $2.5a = 15$

11. $(1/3)x = 9$

12. $n - 1.5 = 4.5$

Two-Step Equations (13-24)

13. $2x + 7 = 19$

14. $3y - 4 = 11$

15. $x/5 + 2 = 6$

16. $4a - 9 = 15$

17. $-2n + 8 = 2$

18. $6x + 1 = 25$

19. $y/3 - 5 = 1$

20. $5 - 3x = -7$

21. $7m + 3 = -11$

22. $x/2 + 8 = 12$

23. $-4a + 10 = -6$

24. $9 - 2y = 3$

Multi-Step Equations (25-36)

25. $3(x + 2) = 15$

26. $2(4y - 1) = 14$

27. $5x + 3 = 2x + 12$

28. $4(a - 3) + 2 = 18$

29. $6n - 4 = 3n + 8$

30. $2(x + 5) - 3 = x + 12$

31. $3(2y - 1) = 5y + 4$

32. $7 + 2(x - 3) = 3x - 1$

33. $4(x + 1) = 2(2x + 3)$

34. $5(a - 2) - 3a = 2a + 1$

35. $3x + 4(x - 2) = 2(x + 4) + 1$

36. $2(3x + 1) - 4 = 5x + 3$

Word Problems (37-40)

37. A number increased by 8, then doubled, equals 30. Find the number.

38. Three times a number minus 7 equals twice the number plus 5. Find the number.

39. A rectangle's length is 5 more than its width. The perimeter is 46. Find dimensions.

40. You have \$45. Books cost \$7 each. How many can you buy and how much is left over?

Section 6: Answer Key

One-Step (1-12)

1. $x=7$ | 2. $y=15$ | 3. $n=7$ | 4. $a=24$ | 5. $x=-5$ | 6. $m=6$ | 7. $x=-7$ | 8. $y=56$ | 9. $x=6.5$ | 10. $a=6$ | 11. $x=27$ | 12. $n=6$

Two-Step (13-24)

13. $x=6$ | 14. $y=5$ | 15. $x=20$ | 16. $a=6$ | 17. $n=3$ | 18. $x=4$ | 19. $y=18$ | 20. $x=4$ | 21. $m=-2$ | 22. $x=8$ | 23. $a=4$ | 24. $y=3$

Multi-Step (25-36)

25. $x=3$ | 26. $y=2$ | 27. $x=3$ | 28. $a=7$ | 29. $n=4$ | 30. $x=5$ | 31. $y=7$ | 32. $x=2$ | 33. No solution | 34. No solution | 35. $x=3$ | 36. $x=5$

Word Problems (37-40)

37. $2(x+8) = 30 \rightarrow x+8 = 15 \rightarrow x = 7$

38. $3x-7 = 2x+5 \rightarrow x = 12$

39. $2(w+5) + 2w = 46 \rightarrow 4w+10 = 46 \rightarrow w = 9, l = 14$

40. $7x \leq 45 \rightarrow x = 6$ books, \$3 left over